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AI-Driven Satellite Image Intelligence for Defense and Coastal Surveillance

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Introduction

Maritime domain plays a vital role in:

- National security
- Global trade and economy
- Environmental monitoring

Large coastal regions like India require efficient surveillance systems

- Satellite imagery provides:
- Wide-area coverage
- Continuous monitoring capability

Artificial Intelligence (AI) enables:

- Automatic detection of objects
- Faster analysis of large-scale data

This project focuses on using AI + Satellite Images for maritime surveillance

Background / Motivation

why we need this?

- Oceans cover over 70% of Earth's surface, making surveillance difficult
- Increase in illegal fishing and maritime smuggling
- Unauthorized ships entering territorial waters
- Traditional systems like radar have limited range
- Need for continuous large-scale monitoring
- AI can analyze satellite images automatically and detect threats quickly

Literature Survey: State of the Art

Object & Ship Detection

Zhangetal.,IEEETGRS2022:Advanced CNNs for maritime vessel identification.

Finding: Achieved 95% accuracy on open-sea detection.

Gap: Lacks robust performance in cluttered coastal zones and varying weather conditions.

Change Detection

Wang&Li,ISPRSJ.2023: Unsupervised methods for land-cover change.

Finding: Detected subtle infrastructure changes with minimal false positives.

Gap: Limited to static changes; fails to track dynamic objects or events.

SAR-Based Detection

Chenetal.,RemoteSens.2021: SAR for all-weather vessel detection.

Finding: Effective through clouds and fog.

Gap: High computational cost and limited feature extraction for small targets.

Domain Adaptation & Operational Systems

Kimetal.,IEEEAccess2023: Transferlearning fornew geographies.

Finding: Reduced training data requirements for new regions.

Gap: Real-time integration into existing defense platforms remains challenging.

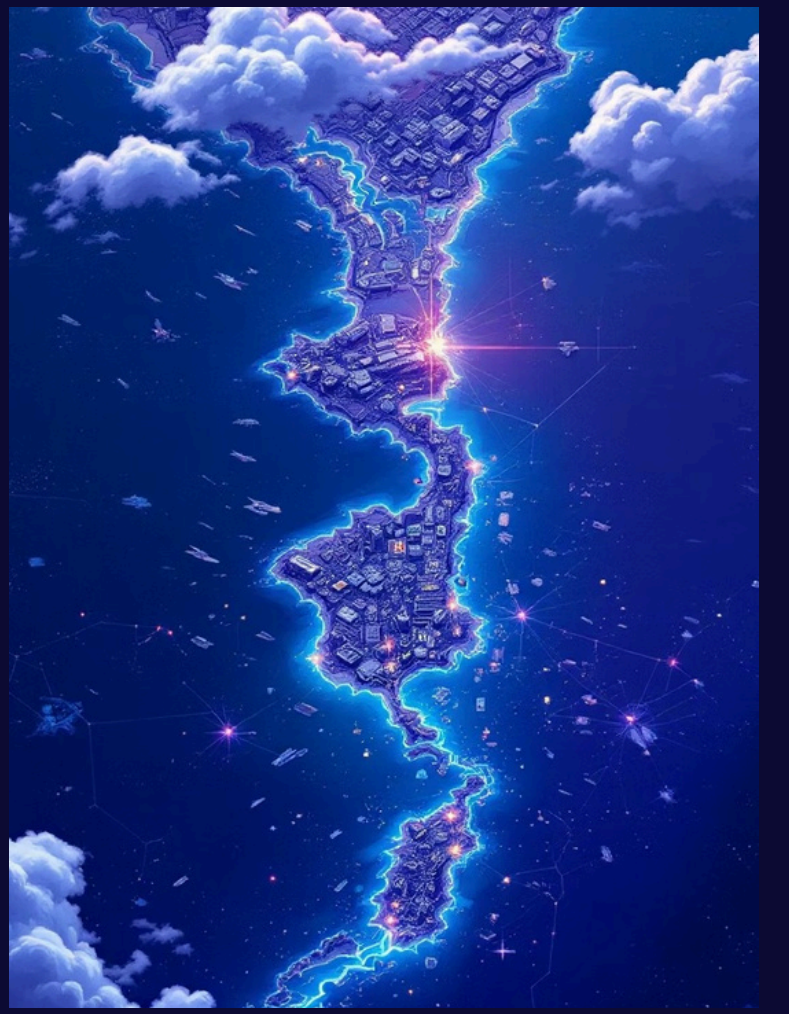
Research Gap

Existing research uses AI techniques for satellite image analysis

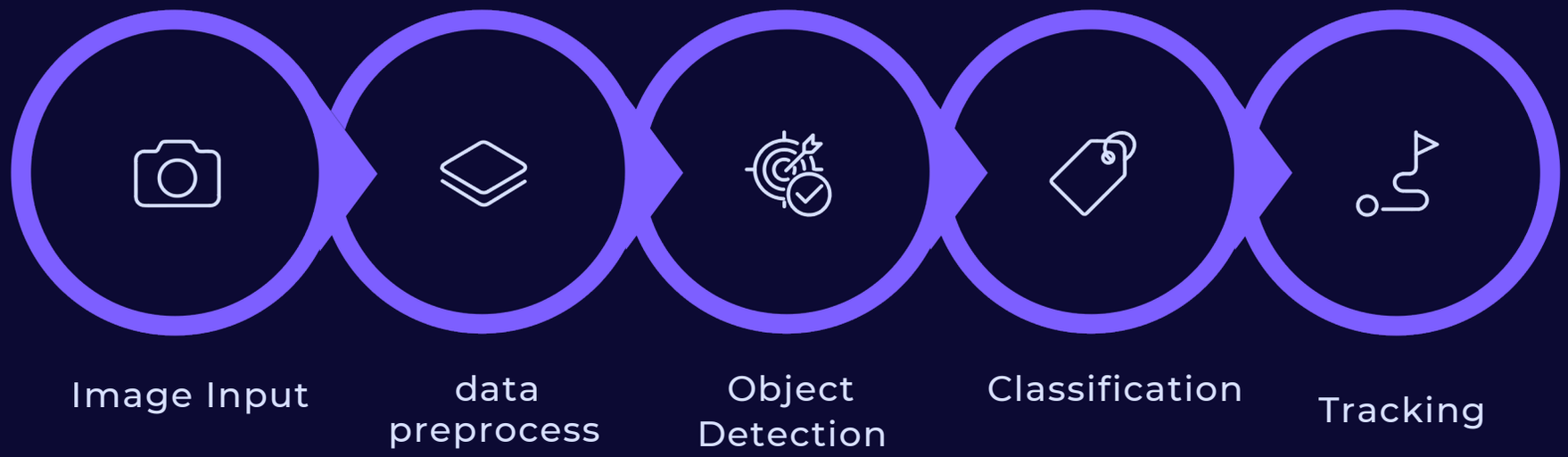
- CNN-based ship detection in satellite images
- YOLO object detection models for maritime surveillance
- Remote sensing techniques for coastline monitoring
- Deep learning models for ocean activity analysis

Gap:

Most systems are expensive, cloud-based, or not offline



AI-Driven Intelligence System Overview



The proposed system integrates satellite data with AI models.

Main components:

1. Satellite image acquisition
2. Image preprocessing
3. AI-based ship detection
4. Threat classification
5. Alert generation system

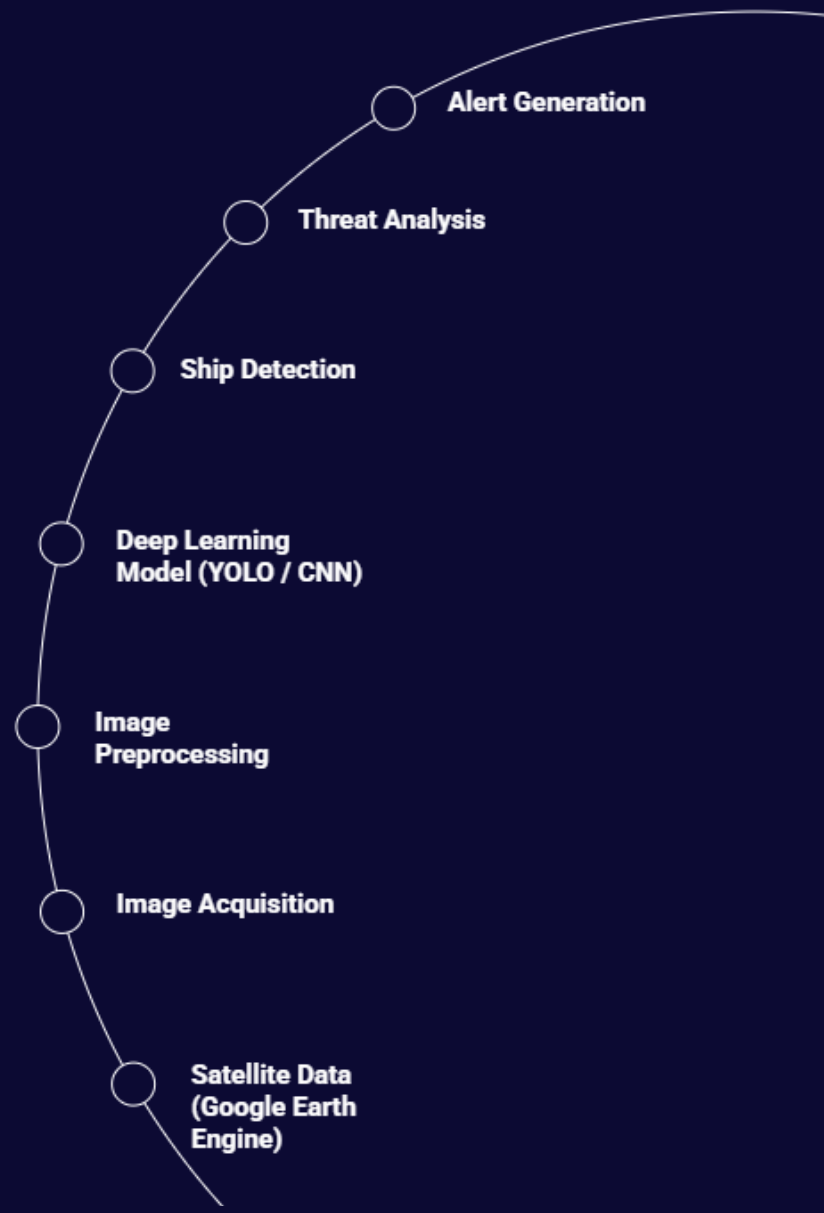
Techniques / Algorithms

System consists of 4 main modules:

- Ship Detection (YOLOv8 + CV Fallback)
- AIS Matching (Spatial Proximity)
- Ship Classification (Size-based)
- Threat Detection (Rule-based Alerts)

Tech Stack:

- Backend: Python (Flask)
- Detection: YOLOv8 + OpenCV
- Frontend: React Dashboard



1. Ship Detection (YOLO + CV Hybrid)

Step 1: YOLO Detection (Primary)

Uses YOLOv8 model
Detects ships with:

- Boundingboxes
- Confidence scores

Step 2: Fallback Detection (If YOLO not available)

Uses Computer Vision:

1. Brightness detection
2. Edge detection (Laplacian)
3. Saturation filtering

Ensures system works offline without model weights

Step 3: Final Processing

Remove duplicate detections (NMS)
Filter noise
Output final ship detections

2. AIS Matching Algorithm

- AIS = Automatic Identification System (ship tracking)
- Each detected ship:
  - Converted to center point
  - Compared with AIS database

Uses distance formula:

- If distance < threshold → AIS matched
- Else → Dark vessel (High risk)

3. Classification & Threat Detection

Ship Classification

Cargo (>4500 px^2)
Patrol (1800–4500 px^2)
Fishing (<1800 px^2)

Threat Levels

Critical → Restricted zone
High → No AIS (dark vessel)
Normal → Safe vessel

Threat Detection Module

Ship Detection & Identification

Automated identification and

classification of maritime vessels, including size, type, and flag state where discernible.

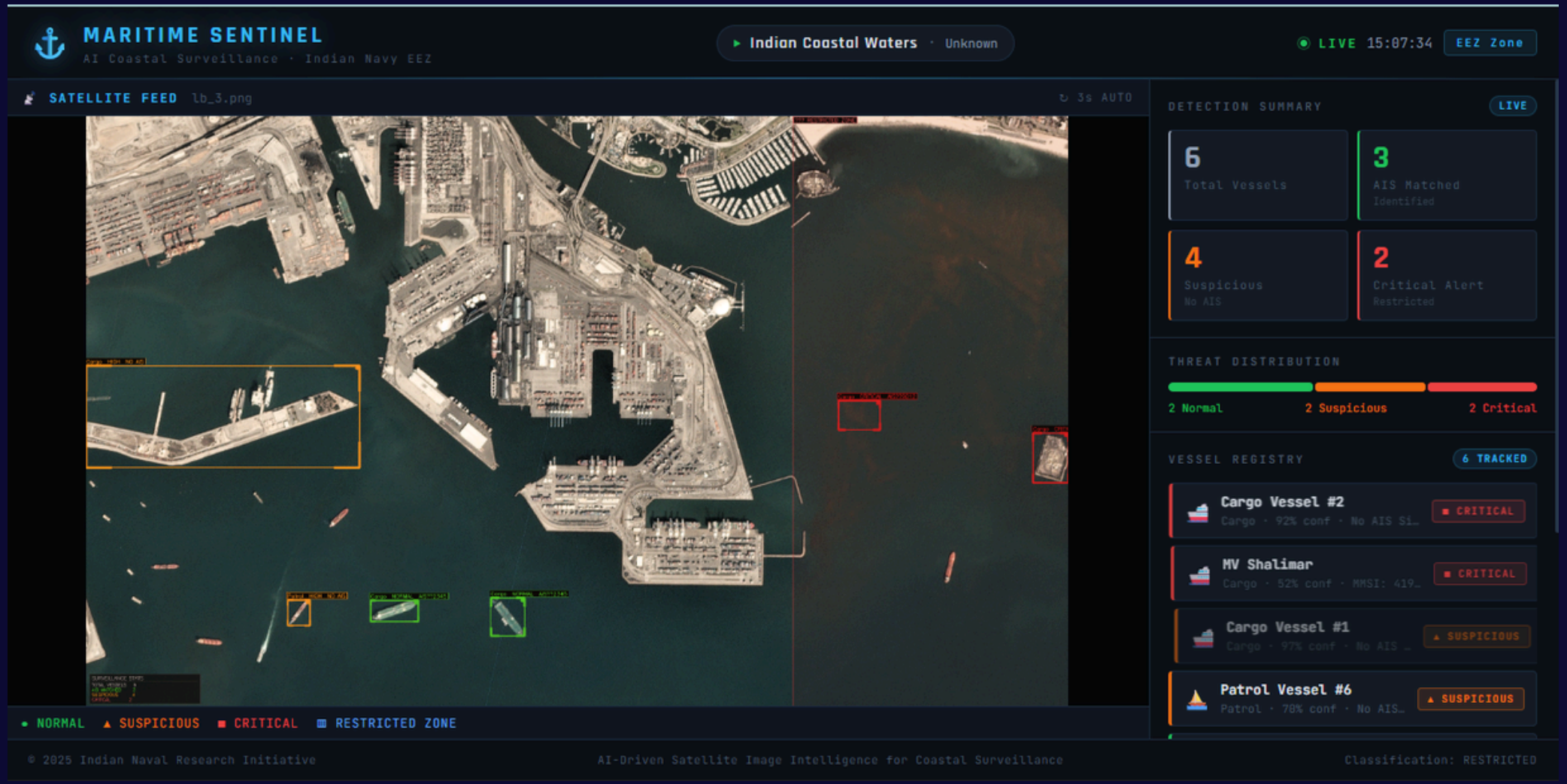
Intrusion Detection

Real-time detection of unauthorized entry into designated exclusion zones or protected areas.

Illegal Activity Recognition

Detection of patternsindicative ofillegal fishing, smuggling, or other illicit maritime operations.

Frontend


MARITIME SENTINEL
AI Coastal Surveillance · Indian Navy EEZ
Indian Coastal Waters · Unknown
LIVE 15:07:34
EEZ Zone
SATELLITE FEED lb_3.png
3s AUTO
DETECTION SUMMARY
LIVE
6
Total Vessels
3
AIS Matched
Identified
4
Suspicious
No AIS
2
Critical Alert
Restricted
THREAT DISTRIBUTION
2 Normal
2 Suspicious
2 Critical
VESSEL REGISTRY
6 TRACKED
Cargo Vessel #2
Cargo · 92% conf · No AIS Si…
CRITICAL
MV Shalimar
Cargo · 52% conf · MMSI: 419…
CRITICAL
Cargo Vessel #1
Cargo · 97% conf · No AIS …
SUSPICIOUS
Patrol Vessel #6
Patrol · 70% conf · No AIS…
SUSPICIOUS
NORMAL
SUSPICIOUS
CRITICAL
RESTRICTED ZONE
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AI-Driven Satellite Image Intelligence for Coastal Surveillance
Classification: RESTRICTED

PROs and CONs

Advantages

- Low cost system
- Automated threat detection
- Faster decision making
- Reduced manual effort

Limitations

- Satellite images may have low resolution
- Weather conditions affect detection

Decision Support & Future Scope

Actionable Dashboards

Intuitive,real-time dashboards presenting threat assessments and geospatial mapping.

Automated Alerts

Instant notificationsf or identified threats, enabling rapid response and mitigation.

Enhanced Defense

Proactivecoastal security, improved maritime domain awareness, and resource optimization.

Future Scalability

Integration with dronefeeds and real-time sensor networks for hyper-local intelligence.